

SyriaTel Customer Churn Analysis Overview

Insights, Modeling, and Strategic
Recommendations

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Customer Churn

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Insights on Customer Retention Strategies

This presentation explores key insights into SyriaTel's customer churn and strategies to enhance retention and improve customer satisfaction.

Overview

Company: SyriaTel, a telecommunications firm operating in Syria.

Issue: Customer retention challenges.

Objective: Discern key indicators signaling potential customer churn.



Churn Insights

Understanding the Factors Leading to
Customer Churn in SyriaTel

Our analysis reveals critical insights into customer
behavior, highlighting key trends that contribute to churn
within SyriaTel's customer base.



Problem Statement

Challenge: Anticipating and comprehending customer attrition to sustain revenue stability.

Goal: Mitigate customer turnover and elevate customer contentment.

Main Objective

Classification Modeling: Analyzing customer churn data to identify key factors influencing customer turnover.

Subjective Objectives

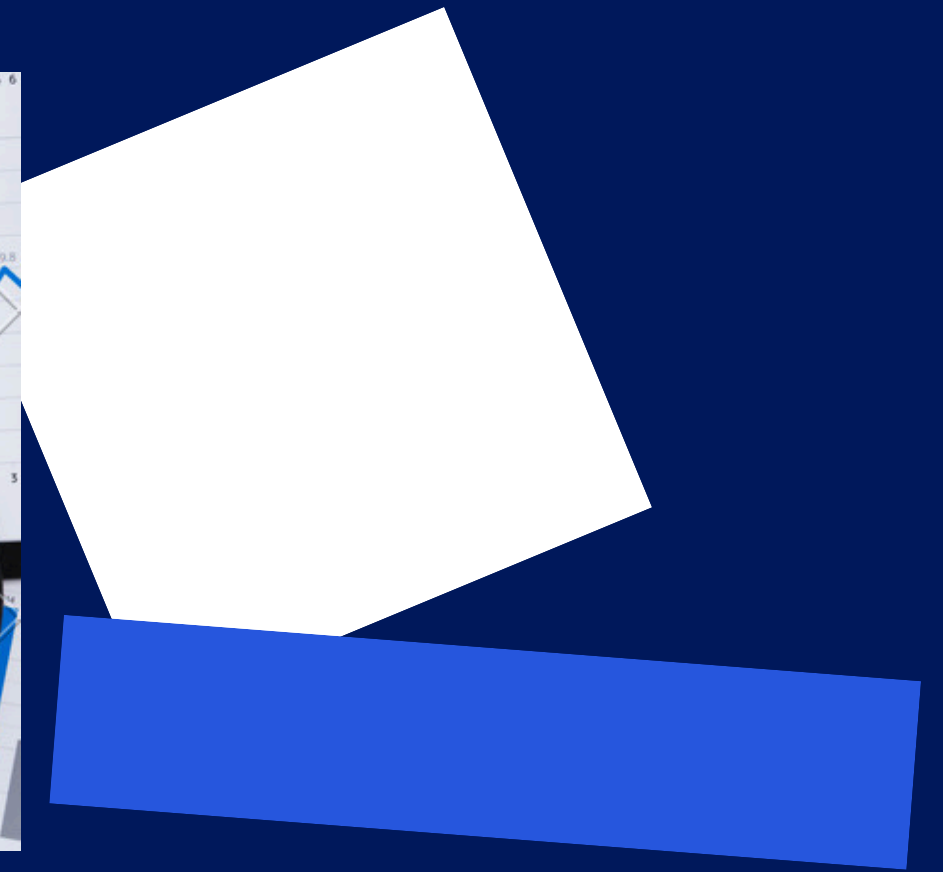
Explore Data: Delve into the correlations between variables and churn rates.

Develop Classification Models: Construct models such as Logistic Regression and Random Forest to predict churn.

Evaluate Feature Significance: Determine the key factors influencing churn prediction.

Assess Model Efficacy: Measure accuracy, precision, recall, and analyze the confusion matrix to gauge model performance.

Propose Recommendations: Deliver actionable insights derived from the model outcomes.



Data Understanding

Analyzing customer churn trends for improved retention strategies and service enhancements.

Baseline Model:

- Accuracy: 87.03%
- Interpretation: The model demonstrates reasonable accuracy by correctly predicting 87 out of 100 instances.

Random Forest Classifier:

- Accuracy: 89.40%
- Best Parameters: max_depth: None, min_samples_split: 2, n_estimators: 200
- Improved Accuracy: 89.80%

XGBoost Classifier:

- Accuracy: 91.24%

Interpretation: This model exhibits the highest accuracy, correctly predicting 91 out of 100 instances.

Conclusions

Insights and Strategies for Retaining Customers



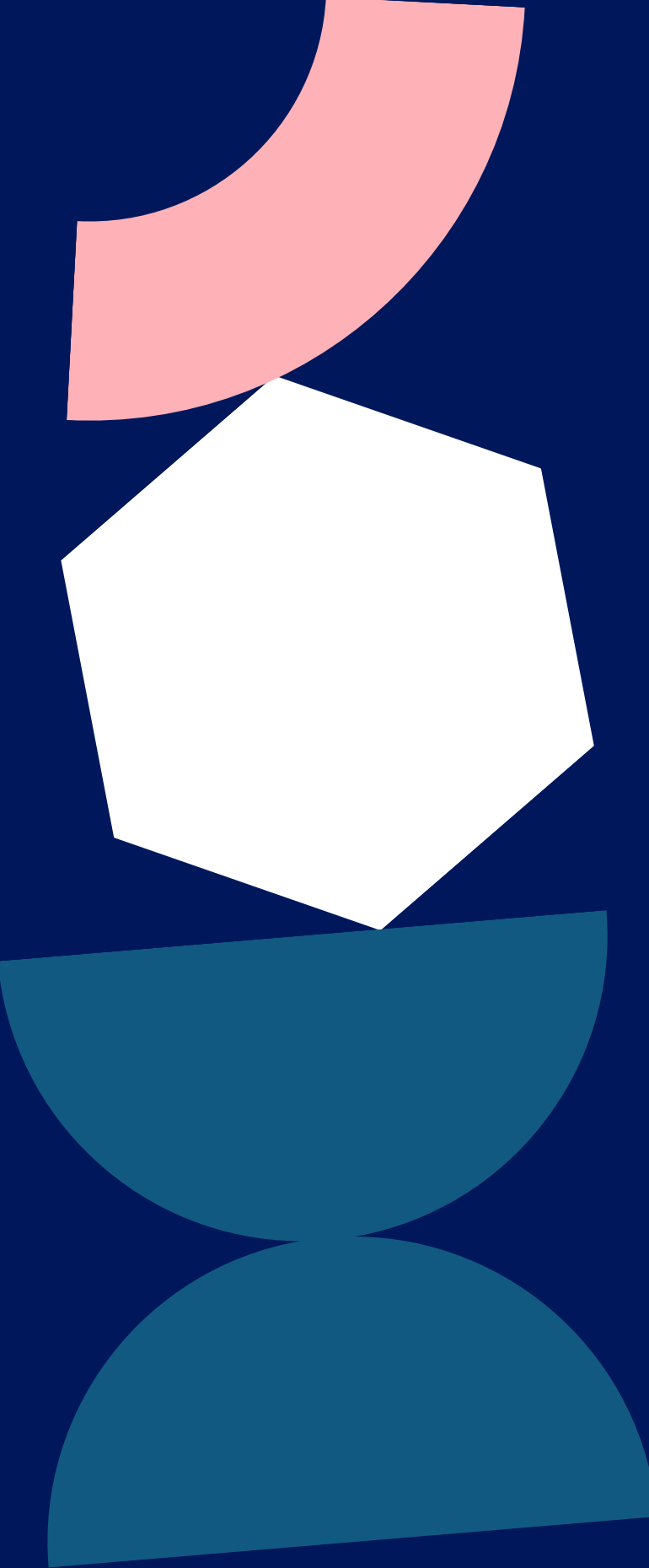
Model Performance: XGBoost achieved an accuracy rate of 91%, followed by Random Forest at 89%, and Logistic Regression at 87%.

Churn Prediction: These models offer valuable insights for identifying customers at high risk of churning.

Key Predictors: Factors such as 'Customer service calls,' 'international plan,' and 'total day minutes' play a significant role in predicting churn.

Recommendations:

1. Targeted Retention Strategies: Deploy customized plans to address high-risk churners.
2. Enhanced Customer Service: Elevate service standards to minimize churn rates.
3. Ongoing Monitoring: Regularly update models with fresh data to ensure accuracy and effectiveness.
4. International Plan Strategies: Develop tailored retention approaches for international plan subscribers.



Questions & Discussion

Open the floor
for audience
questions and
feedback.